# Cyber AI Cup 2026 -- Task 3 Technical Report

### Standalone Vanilla Model Benchmark: LightGBM, XGBoost, and TabPFN v3

### Optuna Hyperparameter Optimization across 5-Fold Stratified Cross-Validation (N = 3,325)

## 1. Executive Benchmark Summary

To establish clean, unencumbered baselines without complex multi-task heads or hierarchical ensembles, we trained and hyperparameter-optimized three standalone vanilla models from scratch on the exact same 63 physical domain features: **Standalone LightGBM** achieved **0.8452 Macro-F1** (84.90% Accuracy), **Standalone XGBoost** achieved **0.8388 Macro-F1** (84.21% Accuracy), and **Standalone TabPFN v3** achieved **0.8108 Macro-F1** (81.14% Accuracy).

## 2. Overall Standalone Model Comparison

| Model Architecture | Macro-F1 | Balanced Acc. | Exact Acc. | Optuna Optimization Details |
|---|---|---|---|---|
| **Standalone Tuned LightGBM** | 0.8452 | 0.8459 | 84.90% | 25 Trials: num_leaves=55, max_depth=11, lr=0.1145, unbalance=True |
| **Standalone Tuned XGBoost** | 0.8388 | 0.8401 | 84.21% | 25 Trials: max_depth=5, n_est=450, lr=0.1019, subsample=0.9218 |
| **Standalone Tuned TabPFN v3** | 0.8108 | 0.8098 | 81.14% | 10 Trials: temp=0.5764, n_est=12, avg_before_softmax=True |

## 3. Detailed 10-Class F1-Score Breakdown

| Traffic Class Name | LightGBM F1 | XGBoost F1 | TabPFN v3 F1 | Support (Nc) | Top Performing Standalone Model |
|---|---|---|---|---|---|
| WhatsApp_voice | 0.9884 | 0.9850 | 0.9648 | 300 | LightGBM (0.9884) |
| WhatsApp_video | 0.9161 | 0.9138 | 0.8639 | 302 | LightGBM (0.9161) |
| GoogleMeet_voice | 0.9007 | 0.9028 | 0.8995 | 300 | XGBoost (0.9028) |
| Discord_video | 0.8906 | 0.8777 | 0.8380 | 456 | LightGBM (0.8906) |
| Messenger_voice | 0.8741 | 0.8571 | 0.8179 | 294 | LightGBM (0.8741) |
| Messenger_video | 0.8675 | 0.8567 | 0.8048 | 331 | LightGBM (0.8675) |
| Discord_voice | 0.8664 | 0.8607 | 0.8039 | 438 | LightGBM (0.8664) |
| Zoom_voice | 0.7654 | 0.7695 | 0.7576 | 300 | XGBoost (0.7695) |
| GoogleMeet_video | 0.7461 | 0.7399 | 0.7363 | 302 | LightGBM (0.7461) |
| Zoom_video | 0.6365 | 0.6252 | 0.6209 | 302 | LightGBM (0.6365) |
| **MACRO AVERAGE** | **0.8452** | **0.8388** | **0.8108** | 3,325 | **LightGBM Winner (0.8452)** |